

DevelopersHub Corporation

AI/ML Engineering – Advanced Internship Tasks

Due Date: 6th October, 2025

Overview

As part of your **AI/ML Engineering Internship** at **DevelopersHub Corporation**, you are required to complete **at least 3 out of the 5 advanced tasks** listed below within the given timeline.

These tasks are carefully designed to give you hands-on experience with **cutting-edge machine learning and artificial intelligence techniques**, including **transformer models, ML pipelines, multimodal learning, conversational AI, and LLM applications**.

You are encouraged to complete all tasks to strengthen your technical portfolio. Technologies and tools you'll use include:

Hugging Face Transformers, scikit-learn, LangChain, Streamlit, Gradio, CNNs, joblib, and LLMs.

Advanced Task Set

Task 1: News Topic Classifier Using BERT

Objective:

Fine-tune a transformer model (e.g., BERT) to classify news headlines into topic categories.

Dataset:

AG News Dataset (Available on Hugging Face Datasets)

Instructions:

- Tokenize and preprocess the dataset
- Fine-tune the `bert-base-uncased` model using Hugging Face Transformers

- Evaluate the model using accuracy and F1-score
- Deploy the model using **Streamlit** or **Gradio** for live interaction

Skills Gained:

- NLP using Transformers
- Transfer learning & fine-tuning
- Evaluation metrics for text classification
- Lightweight model deployment

Task 2: End-to-End ML Pipeline with Scikit-learn Pipeline API

Objective:

Build a reusable and production-ready machine learning pipeline for predicting customer churn.

Dataset:

Telco Churn Dataset

Instructions:

- Implement data preprocessing steps (e.g., scaling, encoding) using **Pipeline**
- Train models like Logistic Regression and Random Forest
- Use **GridSearchCV** for hyperparameter tuning
- Export the complete pipeline using **joblib**

Skills Gained:

- ML pipeline construction
- Hyperparameter tuning with GridSearch

- Model export and reusability
- Production-readiness practices

Task 3: Multimodal ML – Housing Price Prediction Using Images + Tabular Data

Objective:

Predict housing prices using both structured data and house images.

Dataset:

Housing Sales Dataset + Custom Image Dataset (your own or any public source)

Instructions:

- Use CNNs to extract features from images
- Combine extracted image features with tabular data
- Train a model using both modalities
- Evaluate performance using MAE and RMSE

Skills Gained:

- Multimodal machine learning
- Convolutional Neural Networks (CNNs)
- Feature fusion (image + tabular)
- Regression modeling and evaluation

Task 4: Context-Aware Chatbot Using LangChain or RAG

Objective:

Build a conversational chatbot that can remember context and retrieve external information

during conversations.

Dataset:

Custom corpus (e.g., Wikipedia pages, internal documents, or any knowledge base)

Instructions:

- Use LangChain or Retrieval-Augmented Generation (RAG)
- Implement context memory for conversational history
- Retrieve answers from a vectorized document store
- Deploy the chatbot with **Streamlit**

Skills Gained:

- Conversational AI development
- Document embedding and vector search
- Retrieval-Augmented Generation (RAG)
- LLM integration and deployment

Task 5: Auto Tagging Support Tickets Using LLM

Objective:

Automatically tag support tickets into categories using a large language model (LLM).

Dataset:

Free-text Support Ticket Dataset

Instructions:

- Use prompt engineering or fine-tuning with an LLM
- Compare zero-shot vs fine-tuned performance
- Apply few-shot learning techniques to improve accuracy

- Output top 3 most probable tags per ticket

Skills Gained:

- Prompt engineering
- LLM-based text classification
- Zero-shot and few-shot learning
- Multi-class prediction and ranking

Submission Requirements (Per Task)

Each completed task must be **uploaded to your GitHub repository** and submitted via **Google Classroom**.

Checklist:

1. Jupyter Notebook / Script File

- Problem Statement & Objective
- Dataset Loading & Preprocessing
- Model Development & Training
- Evaluation with relevant metrics
- Visualizations (if applicable)
- Final Summary / Insights

2. Code Quality

- Clear structure, logical flow, comments explaining major steps

3. GitHub Repository

- Clear repository name and organization

- **README.md** must include:
 - Objective of the task
 - Methodology / Approach
 - Key results or observations

4. Submission on Google Classroom

- Submit the GitHub repo link for each completed task

Important Notes

- **Deadline:** Complete at least **3 out of 5 tasks** by **6th October 2025**
- You are encouraged to complete all 5 for advanced portfolio building
- Share your work on GitHub and LinkedIn to showcase your skills